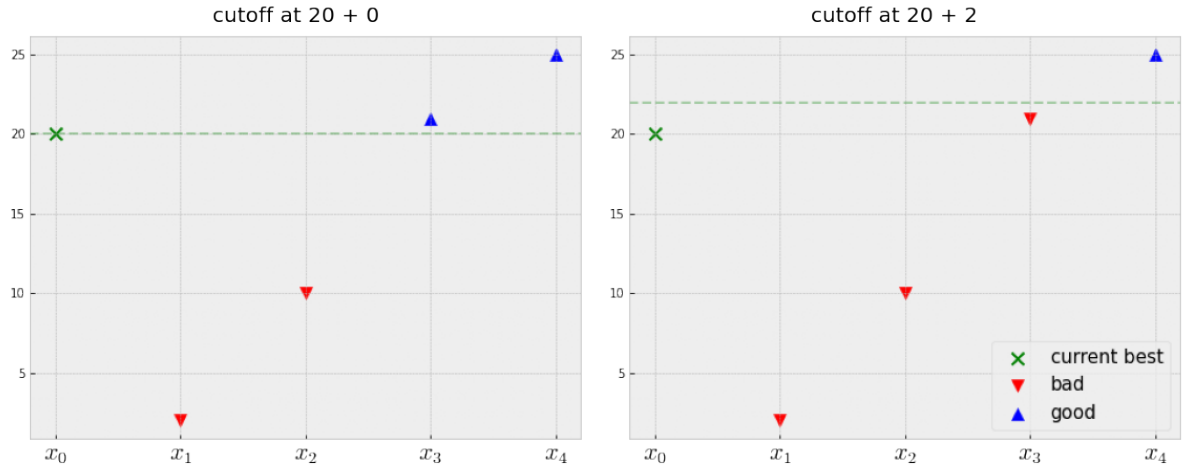


Figure 4.14 Defining a stricter definition of "improvement" by requiring an improvement of at least $\varepsilon = 0$ (left) and $\varepsilon = 2$ (right). The larger the requirement, the more explorative the Probability of Improvement policy becomes. See Exercise 2 for more details.

We won't be looking into more detail here, but Exercise 1 in section 6.2.4 will explore this approach. Interestingly, we will observe that the more improvement we require the PoI to observe, the more explorative the policy becomes.

4.2.4 Exercise 1: Encouraging exploration with Probability of Improvement

One way to address PoI's tendency to over-exploit is to set a higher bar for what counts as "improvement". Specifically, we saw that naïve finding points that maximize the probability of improving from the incumbent prevents us from escaping local optima.

As a solution to this, we can modify the policy to specify that we only accept improvements by at least ε. This would guide PoI to look for improvement in other regions in the search space, once a local region has been sufficiently covered. This exercise, implemented in `CH04/02 - Exercise 1.ipynb`, walks us through this modification and showcases the positive effect it has on PoI. There are multiple steps:

1. Recreate the Bayesian optimization (BayesOpt) loop in `CH04/01 - BayesOpt loop.ipynb`, which uses the one-dimensional Forrester function as the optimization objective.
2. Before the for loop that implements BayesOpt, declare a variable named `epsilon`. This variable will act as the minimum improvement threshold to encourage exploration. Set this variable to 0.1 for now.
3. Inside the for loop, initialize the PoI policy as before, but this time specifying that the incumbent threshold, set by the `best_f` argument, is the incumbent value *plus* the value stored in `epsilon`.
4. Rerun the notebook and observe whether this modification leads to better optimization performance than the original PoI policy by encouraging more exploration.
5. How much more explorative PoI becomes heavily depends on the minimum improvement threshold stored in `epsilon`. Set this variable to 0.001 to observe that a

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